

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

VBCUA is an AI-powered web application that evaluates how effectively users understand and explain concepts through spoken communication. It combines speech transcription, semantic analysis, and fluency assessment into one unified platform, providing students and educators with objective, automated feedback on both concept accuracy and speech delivery.

Project Overview	
Objective	To build an AI-powered tool that evaluates spoken concept explanations by measuring both conceptual understanding and speech fluency, providing automated scores and downloadable feedback reports
Scope	The application covers audio upload, speech-to-text transcription, semantic similarity scoring, fluency analysis, AI-generated feedback, waveform visualization, and PDF report generation — deployed as a public web app
Problem Statement	
Description	Students and learners lack an objective, automated tool to evaluate both the accuracy of their spoken concept explanations and the quality of their speech delivery simultaneously
Impact	Without structured feedback, students cannot identify conceptual gaps or fluency issues, leading to poor interview/presentation performance and inability to track improvement over time
Proposed Solution	

Approach	Combining OpenAI Whisper (transcription), Sentence-BERT (semantic similarity), Librosa (audio feature extraction), and Gemini API (AI feedback) into a unified Streamlit web application
Key Features	Speech-to-text transcription — - Semantic concept scoring — - Fluency analysis (filler words, pause ratio, RMS energy) — - AI-generated feedback — - Waveform visualization — - Downloadable PDF report



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 vCPU (Hugging Face Spaces free tier)
Memory	RAM specifications	16 GB (Hugging Face Spaces free tier)
Storage	Disk space for data, models, and logs	50 GB (Hugging Face Spaces free tier)
Software		
Frameworks	Python frameworks	Streamlit
Libraries	Additional libraries	openai-whisper, sentence-transformers, librosa, torch, fpdf2, matplotlib
Development Environment	IDE	Visual Studio Code
Data		

Data	Source, size, format	User-uploaded audio files (.wav, .mp3, .m4a); Reference concepts stored in JSON (reference_concepts.json)
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